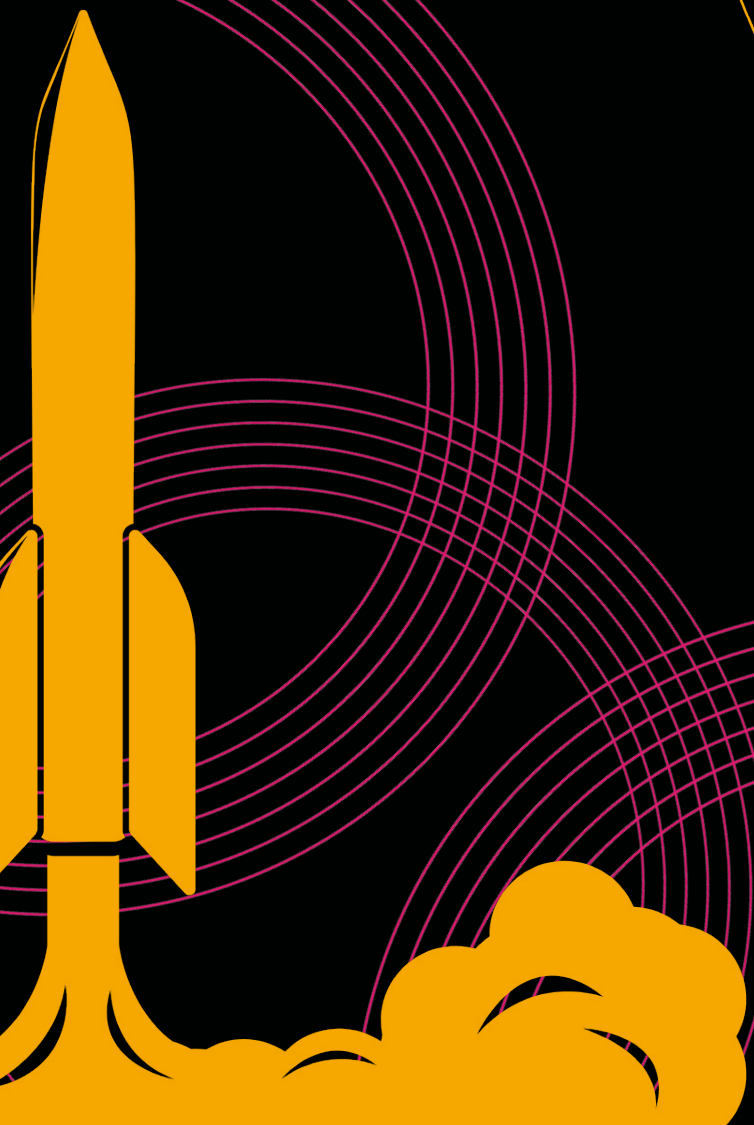
ESPI
European Space Policy Institute

OBSERVATIONS AND NEW PERSPECTIVES FOR EXPLORATION IN TIMES OF CHANGE

SPACE EXPLORATION FOR INNOVATION & COMPETITIVENESS OF EUROPE.



FOREWORD

In March 2023, “Revolution Space”, a report elaborated by an independent High-Level Advisory Group was presented to ESA Council by Anders Fogh Rasmussen former Danish Prime Minister and Secretary General of NATO. It called for Europe to become a leader in space exploration, emphasizing the need for Europe to strengthen its autonomy, resilience, and competitiveness in space. The report warned that otherwise Europe may “fall behind and become dependent of others”.
Like many wake-up calls, that call for a revolution, to ACT NOW remained unanswered. Two years later, reality has overtaken what appeared visionary then. Seismic geo-political and geo-economic shifts put the future of decades of trans-Atlantic partnership and the future of NASA into question. The weakness of Europe’s traditional role as junior partner is fully exposed.

As a first reaction, efforts aiming at the diversification of partnerships takes new prominence, as Europe also turns East. The challenge may turn into a unique opportunity for Europe, and it now is imperative that European policy makers and agencies, together with industry must act to develop the full value of space exploration for innovation and competitiveness of Europe.

1961-1972
NASA’s **Apollo Missions** begin, ushering in a new era of human spaceflight and **exploration** of the Moon.

1998
The **International Space Station (ISS)** is founded as a collaborative project between multiple space agencies to serve as a permanent, orbiting laboratory for scientific research, international cooperation, and preparation for future space **exploration**.

2023
A 12-member “High Level Advisory Group” presented its independent report, ***Revolution Space: Europe’s Mission for Space Exploration***, to the ESA Council, warning that Europe risks strategic and economic dependence if it fails to secure autonomous access to space.

NOW
How will we influence the trajectory of European space **exploration in an uncertain future?**

ABOUT

THIS DOCUMENT IS AN INDEPENDENT ESPI PUBLICATION. IT PROVIDES A NUMBER OF OBERVATIONS AND PERSPECTIVES FOR SPACE EXPLORATION IN TIMES OF CHANGE. IT FOCUSES ON THE ROLE OF EXPLORATION AS A CATALYST OF INNOVATION & COMPETITIVENESS AND FOR ATTRACTING TALENT TO THE SPACE SECTOR. IT IS ELABORATED AS PART OF ESPI’S MANDATE TO PROVIDE INDEPENDENT ANALYSIS, PROPOSALS AND RECOMMENDATIONS ON STRATEGIC SPACE ISSUES TO EUROPEAN DECISION MAKERS AND INSTITUTIONS. IT IS BASED ON STAKEHOLDER CONSULTATIONS, ESPI IN-HOUSE RESEARCH AND IS INFORMED BY RESEARCH CARRIED OUT FOR ESA. IT LEVERAGES ESPI’S EARLIER SUPPORT TO THE WORK ON THE REVOLUTION SPACE REPORT IN 2023. THE QUOTES (❞) IN THIS DOCUMENT WERE COLLECTED DURING INDUSTRY CONSULTATIONS CONDUCTED IN LATE 2024. SIGNIFICANT CHANGES TO THE U.S. SPACE EXPLORATION PROGRAMME SINCE THESE CONSULTATIONS TOOK PLACE RENDER SOME OF THE INDUSTRY COMMENTS EVEN MORE PERTINENT TODAY.

THIS DOCUMENT IS AN INDEPENDENT ESPI PUBLICATION, INFORMED BY RESEARCH CARRIED OUT FOR ESA.

1 POLITICAL DECLINE OF SPACE EXPLORATION IN EUROPE...

Space exploration is the only part of the sector seeing a decline in salience, even though there is a strong increase in political attention towards space in Europe in general.

This indicates that the policy value of space exploration is not understood at a political level in Europe.

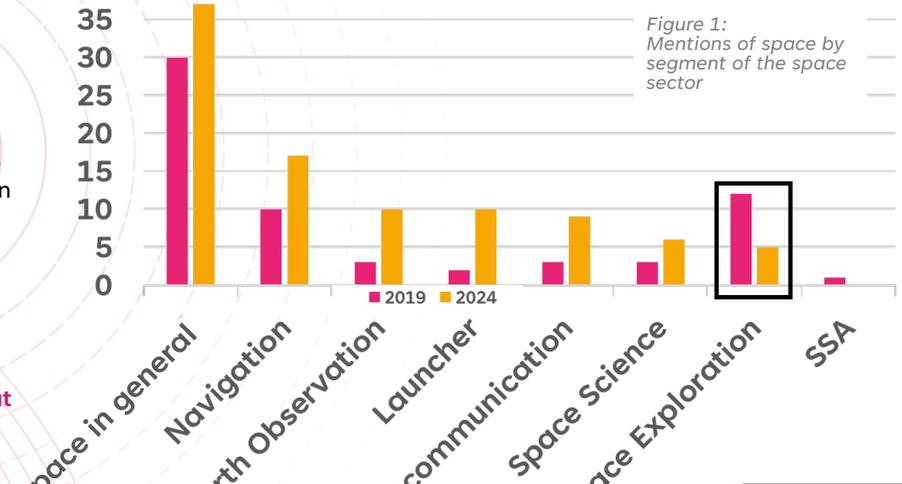


Figure 1: Mentions of space by segment of the space sector

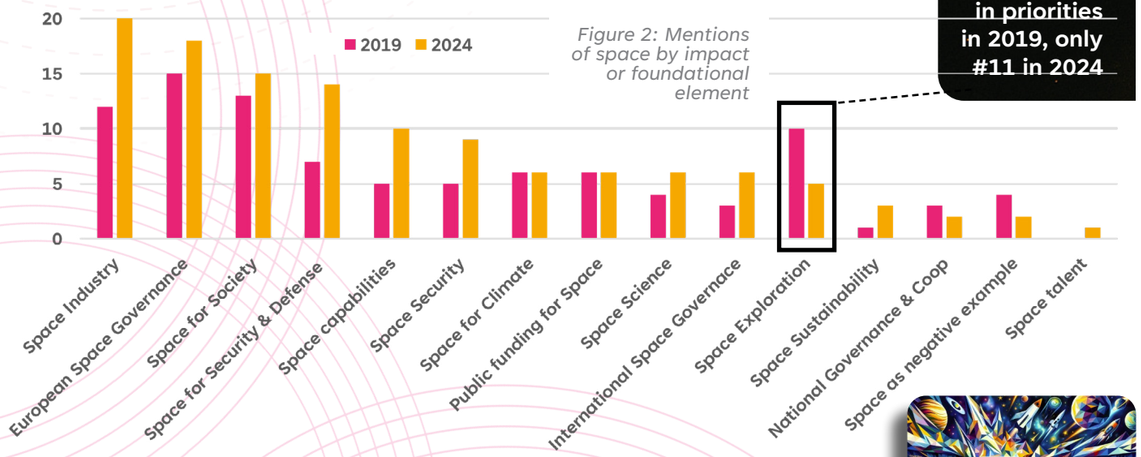


Figure 2: Mentions of space by impact or foundational element

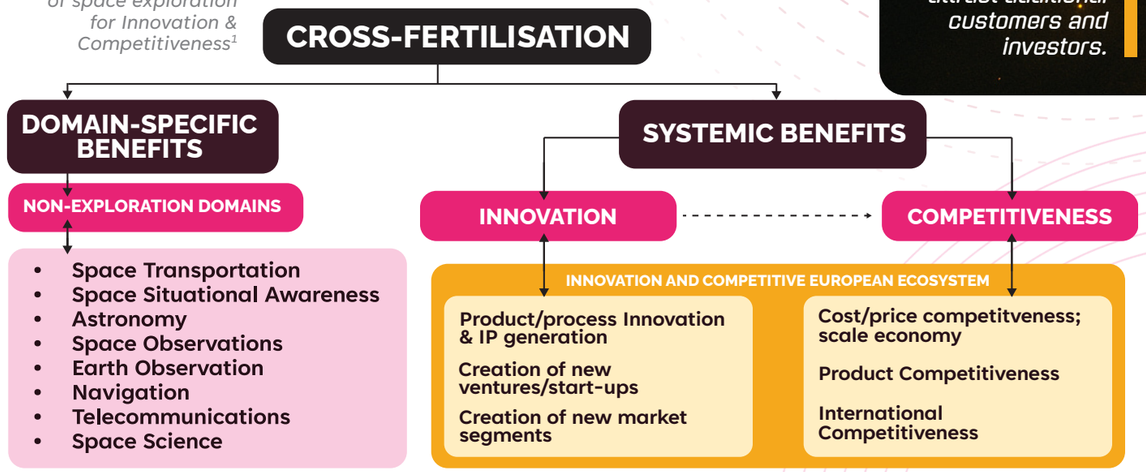
Exploration ranked #4 in priorities in 2019, only #11 in 2024

Political attention is measured through mentions of space in party manifestos in regard to the European Parliament Election 2019 and 2024. More info can be found in **"Will space sit in the European Parliament?"**

2 ...YET SPACE EXPLORATION CREATES VALUE FOR EUROPEAN INNOVATION AND COMPETITIVENESS

The role of ESA as enabler to develop new services, as well as to act as institutional customer is key to attract additional customers and investors.

Figure 1: The benefits of space exploration for Innovation & Competitiveness[1]



Cross-fertilisation from exploration operates on two fronts:

- **at the domain-specific level**, spillover benefits brought by exploration activities or technologies in space programmes, and
- **at the systemic level**, generating benefits on the innovativeness and competitiveness of the broader European space ecosystem.

IMPACT CASE 1: In-space additive manufacturing

Given the different technological requirements for operating in microgravity, human presence in LEO requires different approaches to problem-solving which are true for the broader space sector. Specific capabilities, such as in-space additive manufacturing, which originated as a support capability to assist astronauts in daily tasks, now have technological implications that could help reduce launch costs and revolutionise in-orbit assembly and manufacturing. For instance, breakthrough technologies such as the Additive Manufacturing Facility built by Made In Space (now Redwire), which were tested and operated aboard the ISS marked an important milestone for ISAM and exploration as a whole.

[1] This assessment is focused on the impact of exploration for the Innovation & Competitiveness of the space sector and excludes the manifold benefits that exploration brings to terrestrial sectors like pharma, agriculture, automotive etc. This analysis is also not focused on the direct, indirect, induced & catalytic benefits, which have been explored by ESPI and BCG in "More than a Space Programme"

IMPACT CASE 2:
In-orbit refuelling

Exploration-enabled technologies, like in-orbit refuelling, hold transformative potential for the space economy. By leveraging platforms like the ISS, commercial actors, especially in the US, not only validate their technologies but also position themselves to capture emerging market segments. The growing collaboration between U.S. and European stakeholders, exemplified by Orbit Fab's integration into European operations, underscores exploration's far-reaching economic impacts across the global space sector, which is not always fully captured by European companies.

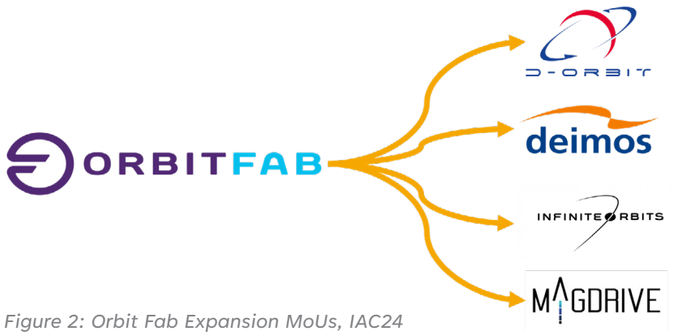


Figure 2: Orbit Fab Expansion MoUs, IAC24

IMPACT CASE 3:
International Competitiveness

International cooperation in exploration, historically with NASA, was the determining factor of ESA's exploration heritage. The construction of exploration modules such as SPACELAB and MPLM was key in defining and nurturing European industry's technical expertise and played an important role in maintaining a European foothold in the US market until now. This critical juncture is what so far kept European industrial players attractive contractors for US exploration and non-exploration initiatives. Both Airbus and Thales have drawn significant profits from exploration programmes, representing 15% of their annual revenue in 2022.[2] Although their status as European champions and their large involvement in ESA programmes helps them maintain their status quo, it's this **technical expertise that has thus far allowed them to win US commercial contracts outside of ESA-NASA collaborations.**

Figure 3: European OEM Participation in ESA NASA Collaborations

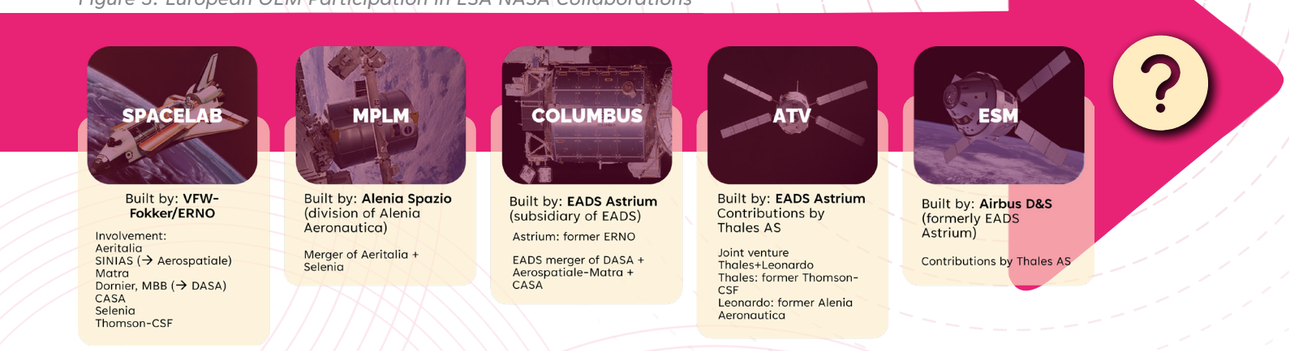


Figure 4: From ESA-NASA to international commercial programmes

[2] ESPI/BCG: More than a Space Programme (Link)

3 EUROPE DOES NOT CAPTURE THE TRUE INNOVATION & COMPETITIVENESS BENEFITS OF EXPLORATION

There is a significant gap between how public investment in exploration has driven innovation and competitiveness in places like the U.S., compared to Europe.

	GLOBAL RATING (PAST ACTIVITIES)	ESA EXPLORATION RATING
INNOVATION		
Product/Process Innovation & IP Generation: Emphasis on developing new products or processes and generating intellectual property	4.9	4.1
Creation of New Ventures: Focused on the establishment of new enterprises, particularly startups	4.8	3.8
Creation of New Market Segments: Targeted the expansion into new customer segments	4.6	4.0
COMPETITIVENESS		
Cost-Price Competitiveness: refers to a company's ability to offer products or services at lower prices than its competitors while maintaining acceptable quality levels. This is often achieved through economies of scale and low-cost inputs	4.2	3.1
Product Competitiveness: Refers to the capacity to produce and bring R&D to Market as well as Access to risk capital	4.8	3.8
International Competitiveness: refers to a company's ability to effectively access foreign markets and capture market shares in those markets	4.6	3.3

Figure 5: Rating of Indicators' performance for global and ESA exploration initiatives. (1-6 scale)

Specifically, **Europe has been unable to translate the innovation benefits stemming from exploration into a robust competitive edge.** Industry representatives highlighted challenges related to **cost and price competitiveness**, particularly in comparison to the U.S. market. Respondents noted that Europe lags behind the United States by an estimated decade in private sector competitiveness, which underscores the need for a comprehensive, long-term investment strategy, incentivising private investment, e.g. through anchor customer contracts.

> *The target shall not be only to diversify market segments inside the space sector as this risks splitting the overall space budget into too many small segments without a critical mass. It shall rather be used to open up new non-space market segments that benefit from Space.*

> *Gap with US market is enormous and difficult to be covered. Investment in space exploration (private/public) are crucial to build-up a Cost/Price Competitive industry as US situation demonstrated clearly (e.g. SpaceX, Blue Origin, etc). In EU a more incisive long-term plan need to established and fostered to increase Competitiveness in private sector (as of today, at least 10 years in delay with regard to the US)*

> *Economies of scale can only be realized if Europe buys European. However, lowest prices in the supply chain are not always European, causing a delicate balancing issue between strategic and commercial interests.*

Inputs from industry in the consultation campaign

4 SPACE EXPLORATION DRIVES THE INSPIRATIONAL POWER OF SPACE TALENTS

All data retrieved from a survey for the ESPI Talent Study to be released Q3 2025

Inspiration stands out as the top motivator for those pursuing a career in the space sector. Over 96% of respondents cited inspiration as one of the key reasons driving their ambition to enter the field.

Inspirational factors can be grouped into two categories: **space exploration and sciences** and **concrete space applications on earth**.

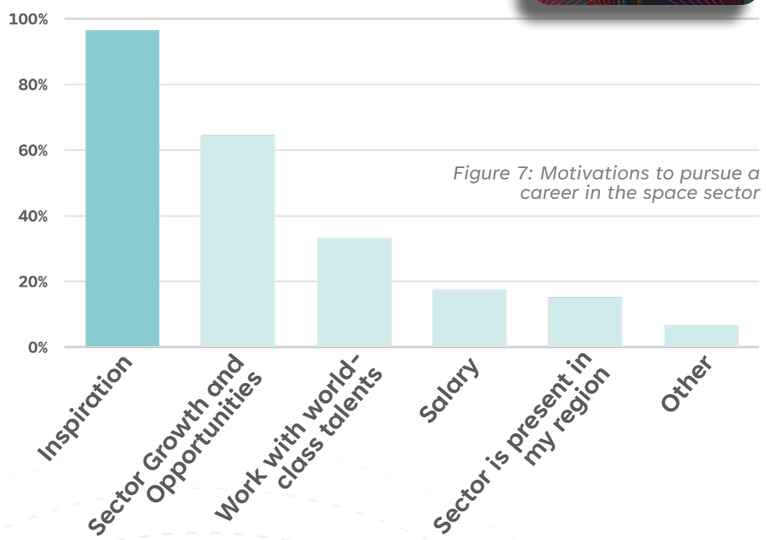


Figure 7: Motivations to pursue a career in the space sector

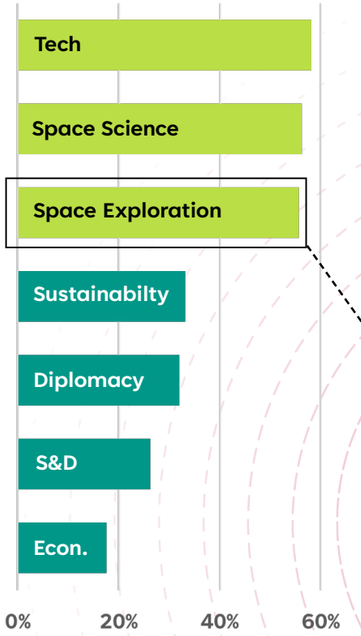


Figure 8: Share of respondents inspired by inspirational factors

Space Exploration (together with Space Science and Technology) are the **main factors contributing to the inspirational power of the space sector,** while the more concrete space applications on earth (economy, security, climate, diplomacy) fare significantly lower in comparison. About 55% of respondents that are inspired by space in general, are inspired or interested by the former, while only about 30% have a fascination for the latter.

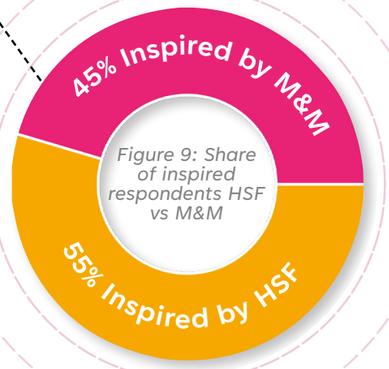


Figure 9: Share of inspired respondents HSF vs M&M

Within Space Exploration, both elements, Human Spaceflight **(HSF)** and Moon & Mars exploration **(M&M)** have a comparable inspirational effect.

EXPLORATION DRIVES A LOYAL, COMMITTED, AND PURPOSEFUL WORKFORCE

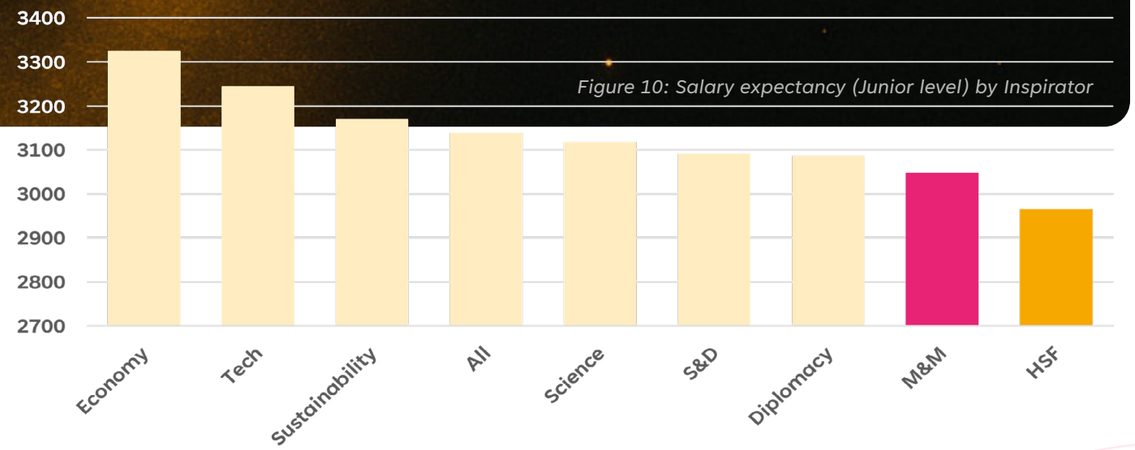


Figure 10: Salary expectancy (Junior level) by Inspirator

The captivating pull of space exploration in retaining individuals within the space sector is reflected indirectly in their salary expectations and willingness to also work in other industries. Inspiration plays a stronger role in attracting people to the space sector than the direct monetary compensation. This relationship is strongly influenced by the inspirational power of space exploration specifically. **Students and young professionals inspired by Exploration have significantly lower salary expectation** for junior positions than the ones driven by other parts of the space domain. This lower salary expectations hints that **space exploration creates a stronger internal motivation and purpose** for young talent to work in the space sector compared to other domains.

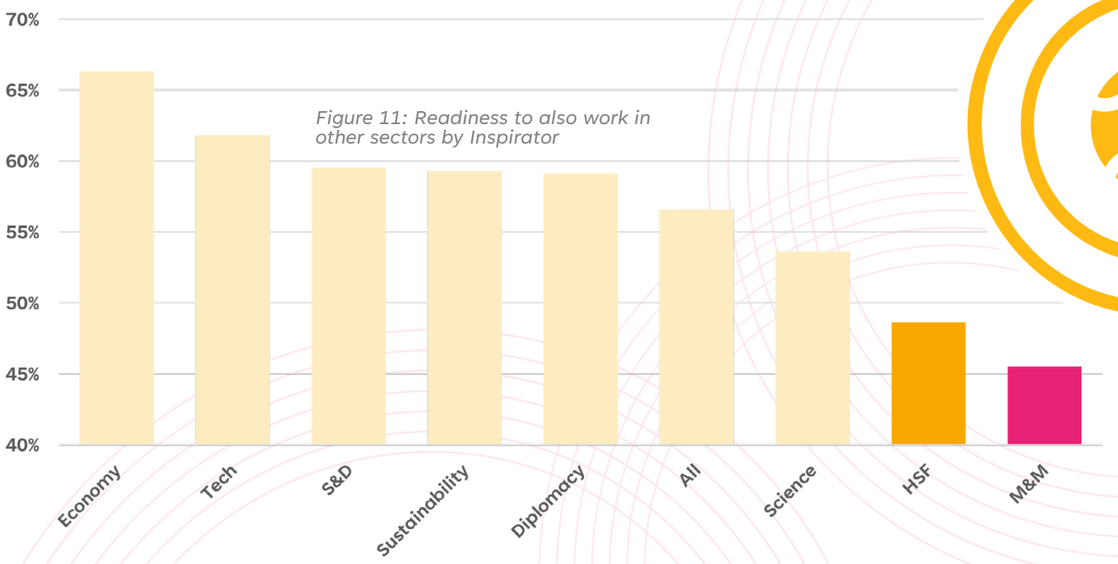


Figure 11: Readiness to also work in other sectors by Inspirator

Additionally, **people who are inspired through exploration seem to generally be more loyal and committed to the space sector.** The share of people inspired through either HSF or M&M, who can imagine also working in other sectors, is significantly lower than for other, more earth-related inspirators, such as economy, sustainability, or security. These individuals are less likely to apply to jobs outside the space sector.

5 INDUSTRY EXPECTS A MORE AUTONOMOUS EUROPE IN EXPLORATION

Already by the end of 2024, a large portion of the consulted European space exploration industry expected the Trump administration to push **Europe toward a "Symphonie Moment", (an autonomous European approach to exploration).**

Symphonie was Europe's first communication satellite project. The two geostationary satellites composing Symphonie were built in the early 1970s by a Franco-German consortium and meant for commercial exploitation. The initial US refusal to provide the launch service if the satellites were used for commercial purposes made it clear that Europe had to free itself from US dependence in accessing space. While the United States eventually launched the two satellites in 1974 and 1975 under the restrictive clause of an exclusively experimental use, the quest for European autonomy was the key driver behind the development of the Ariane launcher and a competitive satcom sector.

Figure 6: European industry's perceptions regarding the impact of Trump administration

Without strategy - Trump presidency will ignore Europe in Space Exploration and take advantage of our know how by attracting directly our talents.

Trump will foster exploration, but more for US leadership rather than a soft power diplomacy tool.

Less opportunities for Europe to support US programs, leading to less investment in space in Europe and lost of competitiveness. Europe should not only look west (e.i. the US) for international collaboration but also look east (Middle East, India, Japan). Even if the US are a key partner, Europe should work with all space agencies in the world and be the link between all of them. That's where Europe is strong.

Europe will need to technically compare American abilities and their speed in technology progress to be competitive.

US investments in Exploration will rise, competition will be even more aggressive. European investments must rise as well to gain an Autonomy in exploration and protect security, capabilities and industry

America First approach will imply a reinforced role of US companies in the International Projects. ESA and European Industry shall work in order to develop capabilities and a sustainable supply chain decoupled with regard to the american market.

Trump will adopt a "US-first" approach to internal investment and international cooperation.

Europe needs to take control over its own roadmap and push programs it can finance. Robotic exploration and development of applications for Pharma/Biotech can be THE major next market in which Europe can excel and is able to finance a strategy of non-dependance.

6 RECOMMENDATIONS FOR EUROPE TO SCALE UP ITS AMBITION IN SPACE EXPLORATION AS A UNIQUE WINDOW OF OPPORTUNITY FOR THE CONTINENT

Promote the policy value of exploration to policymakers across Europe

Implement a truly autonomous approach to LEO activities, with a Europe-owned infrastructure, FOR **A STRONG EUROPE...**

Pursue stronger cooperation and diplomatic engagement with actors such as India, Japan or South Korea, and also non-space sectors such as pharma and food industries. **... AS PARTNER TO THE WORLD**

Recognise the full potential of space exploration in attracting, developing, and retaining talent for the space sector, and establish a dedicated talent development component within the exploration programme.

Boost creation of new segments (e.g. IOS)

Fully recognise the serendipitous benefits brought by exploration

Access of European industry to foreign markets

Double down on Innovation

Fuel Competitveness

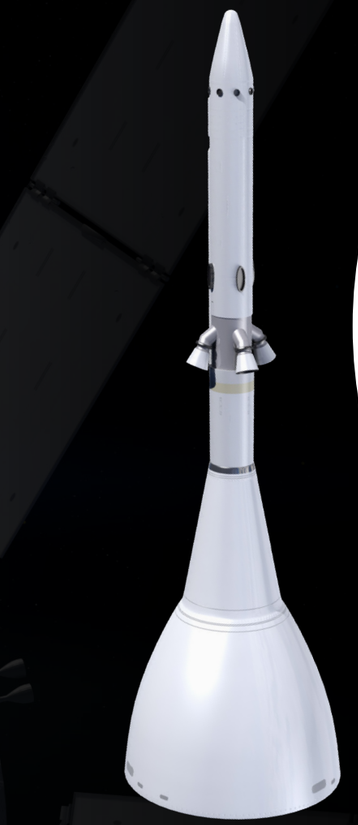
Establish a programme simultaneously acting on domain-specific benefits & systemic benefits of the broader European ecosystem.

Beyond that, collaboration with non-space sectors that benefit from exploration infrastructure (such as pharma and food industry) should be established / strengthened. In that sense, ESA should adopt an ambidextrous approach —an organizational strategy that balances exploitation (LEO) and exploration (Moon).

7 THE FUTURE IS UNCERTAIN FOR THOSE NOT CRAFTING IT

In a volatile international context where space exploration serves political and strategic agendas, the cost of serving as a junior partner and overreliance on partners outweighs the necessary investment in one's own capabilities, to make Europe a more autonomous and consequently also stronger partner to the world.

Photo: ©NASA



Orion's Sevice Module, also known as the European Service Module, was designed through the **collaborative efforts of ESA and NASA.** It provides Orion's crew access to vital navigation, propulsion and life support mechanisms.

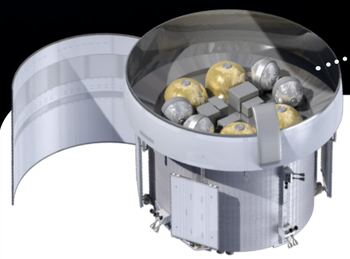
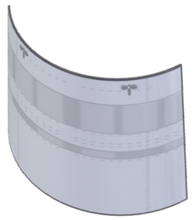


Photo: ©NASA



Space exploration is a strategic asset that can shape global influence and alliances. Bringing its value to the forefront of attention of European policymakers can empower a more effective policy and space diplomacy.

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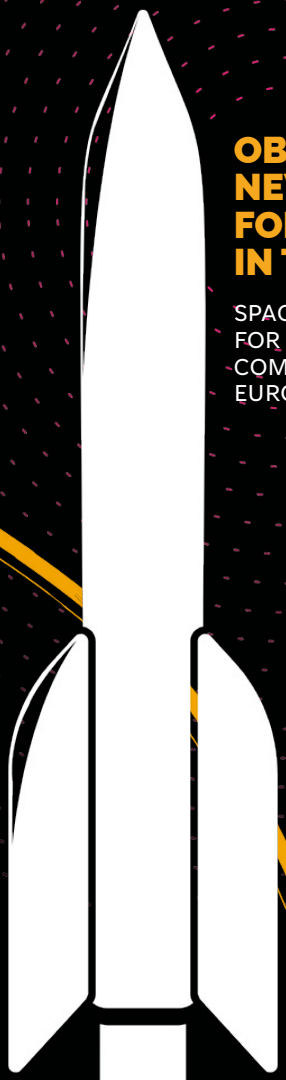

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